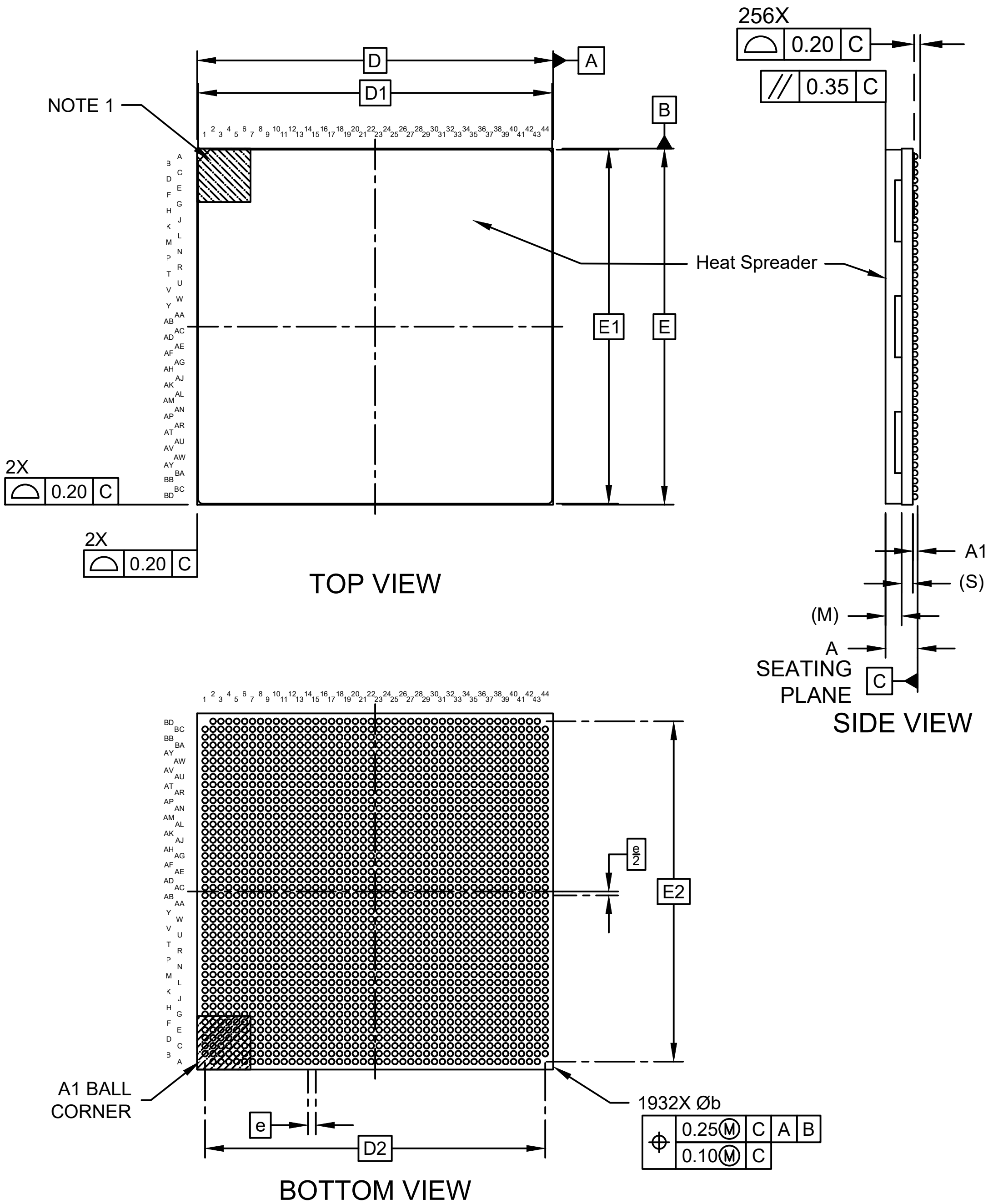
1932-Lead Very Thick Ball Grid Array (6WW) - 45x45x4.28 mm Body [HB2BGA] With Heat Spreader

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



Microchip Technology Drawing C04-00636 (6WW) Rev B Sheet 1 of 2

1932-Lead Very Thick Ball Grid Array (6WW) - 45x45x4.28 mm Body [HB2BGA] With Heat Spreader

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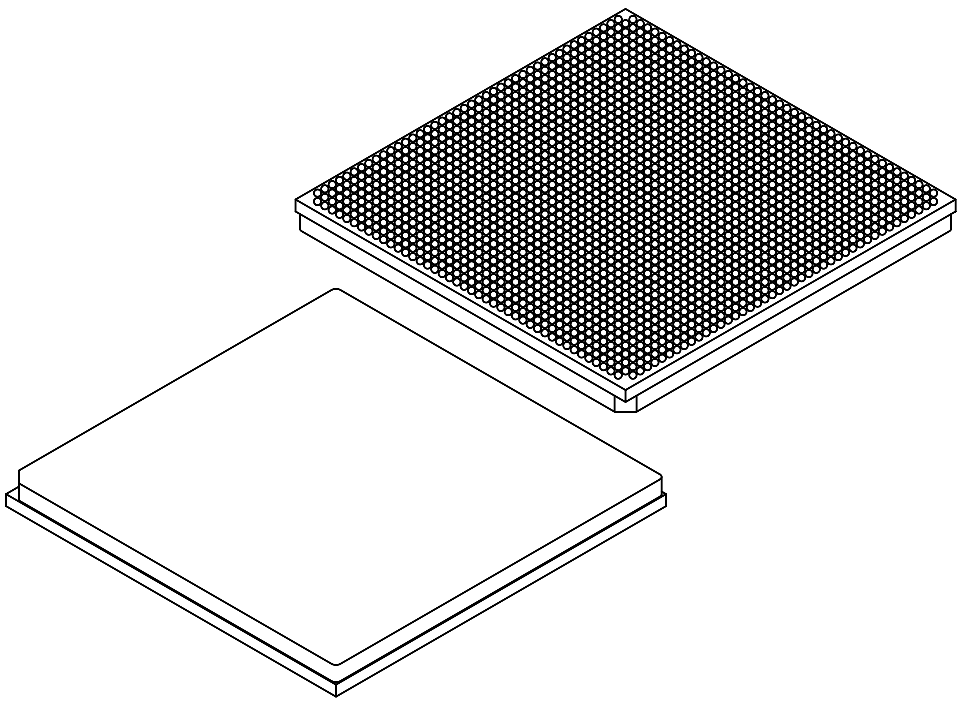


Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N		1932	
Pitch	e		1.00 BSC	
Overall Height	A	3.82	4.05	4.28
Ball Height	A1	0.40	0.50	0.60
Heat Spreader Thickness	M	–	2.13 REF	–
Substrate Thickness	S		1.42 REF	
Overall Length	D		45.00 BSC	
Heat Spreader Length	D1		44.75 REF	
Ball Array Length	D2		43.00 BSC	
Overall Width	E		45.00 BSC	
Heat Spreader Width	E1		44.75 REF	
Ball Array Width	E2		43.00 BSC	
Ball Diameter	b	0.50	0.64	0.70

Notes:

1. The Pin 1 visual index feature may vary, but it must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M

   BSC: Basic Dimension. The theoretically exact value is shown without tolerances.

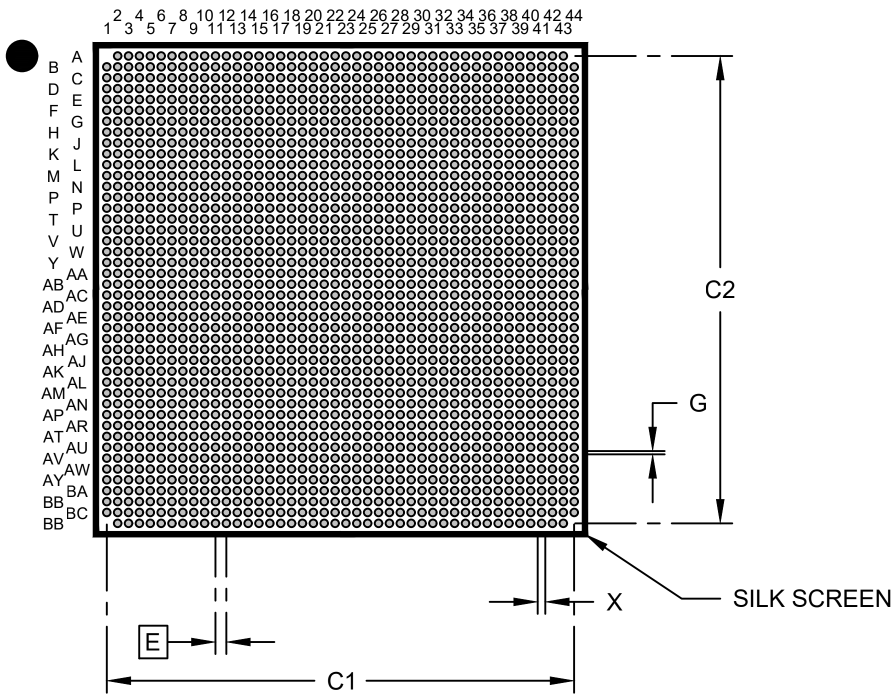
   REF: Reference Dimension, usually without tolerance, is for information purposes only.

Microchip Technology Drawing C04-00636 (6WW) Rev B Sheet 2 of 2

Land Pattern

1932-Lead Very Thick Ball Grid Array (6WW) - 45x45x4.28 mm Body [HB2BGA] With Heat Spreader

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

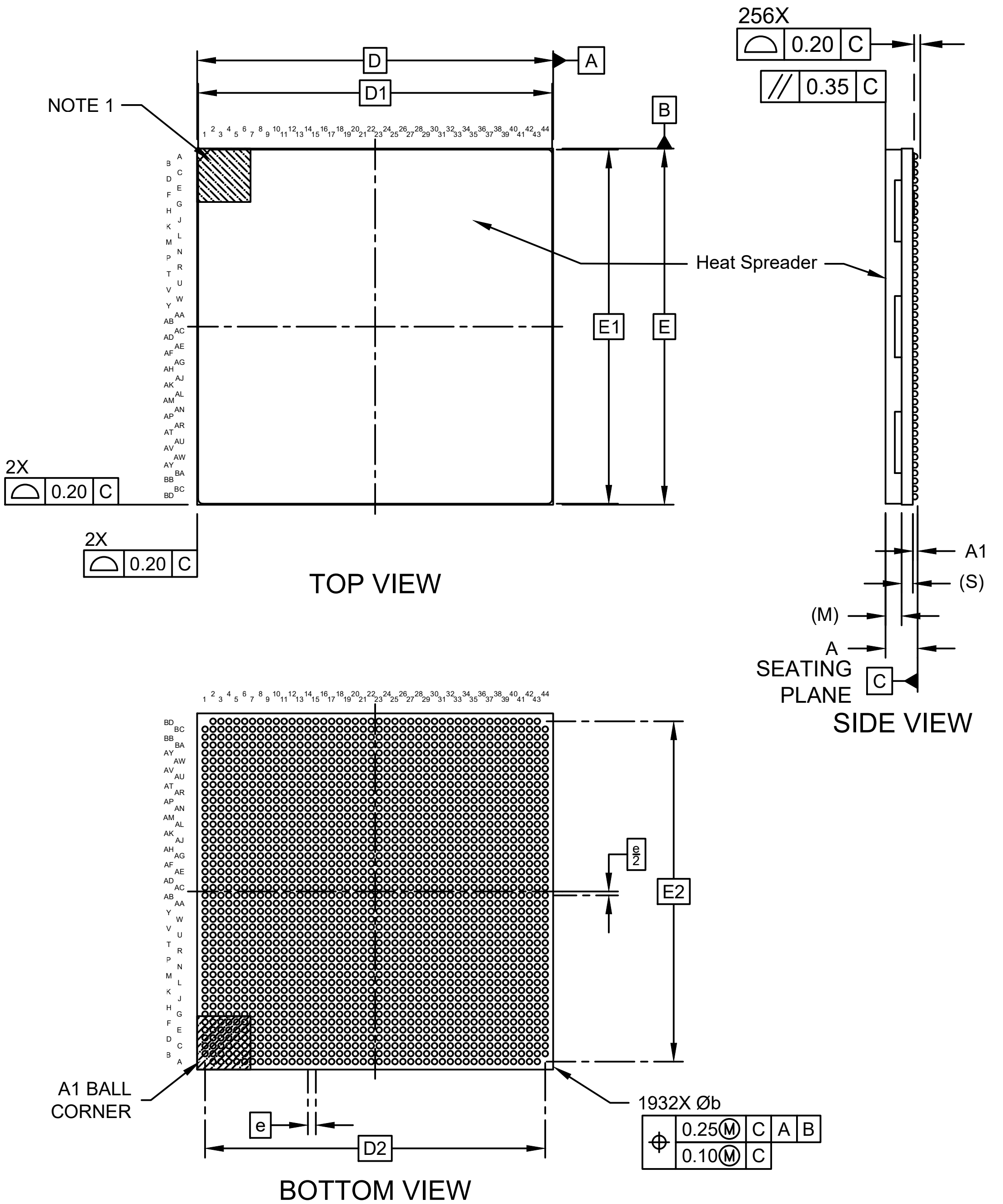
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	1.00 BSC		
Contact Pad Spacing	C1		43.00	
Contact Pad Spacing	C2		43.00	
Contact Pad Diameter (X1932)	X			0.55
Contact Pad to Contact Pad	G	0.30		

Notes:
1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. The theoretically exact value is shown without tolerances.
2. For best soldering results, pleese refer to the current industry standard IPC-7093.

Microchip Technology Drawing C04-02636 (6WW) Rev B

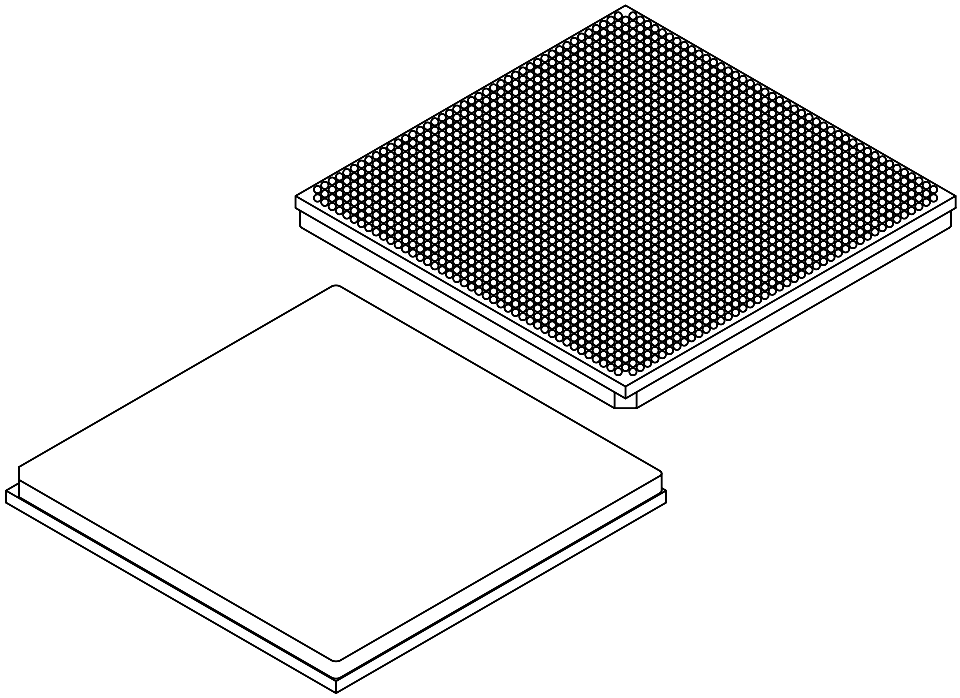
1932-Lead Very Thick Ball Grid Array (9AW) - 45x45x4.28 mm Body [HB2BGA] With Heat Spreader

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



1932-Lead Very Thick Ball Grid Array (9AW) - 45x45x4.28 mm Body [HB2BGA] With Heat Spreader

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N		1932	
Pitch	e		1.00 BSC	
Overall Height	A	3.82	4.05	4.28
Ball Height	A1	0.40	0.50	0.60
Heat Spreader Thickness	M	–	2.13 REF	–
Substrate Thickness	S		1.42 REF	
Overall Length	D		45.00 BSC	
Heat Spreader Length	D1		44.75 REF	
Ball Array Length	D2		43.00 BSC	
Overall Width	E		45.00 BSC	
Heat Spreader Width	E1		44.75 REF	
Ball Array Width	E2		43.00 BSC	
Ball Diameter	b	0.50	0.64	0.70

Notes:

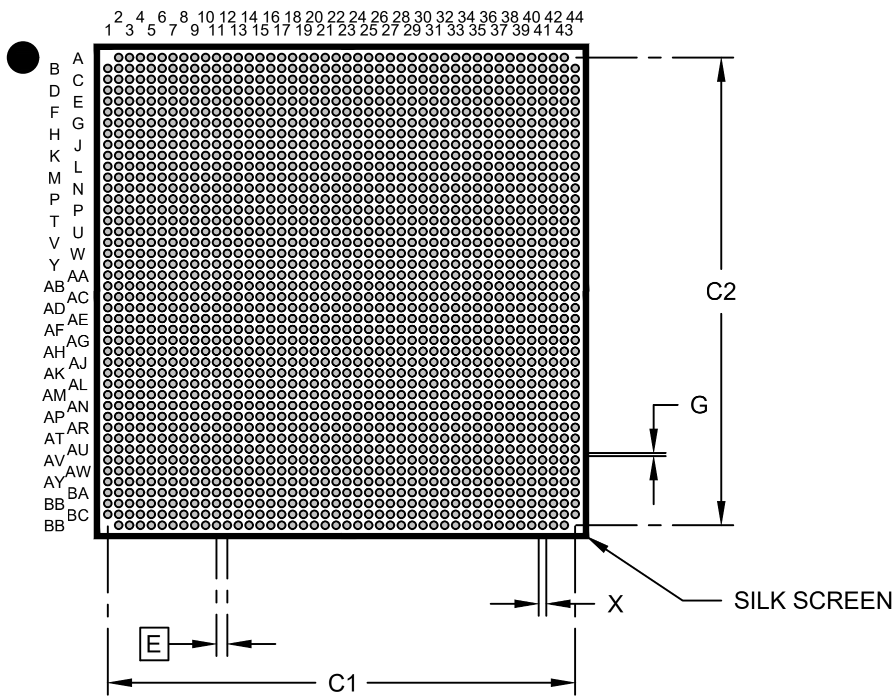
1. The Pin 1 visual index feature may vary, but it must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. The theoretically exact value is shown without tolerances.
   REF: Reference Dimension, usually without tolerance, is for information purposes only.

Microchip Technology Drawing C04-00636 (9AW) Rev B Sheet 2 of 2

Land Pattern

1932-Lead Very Thick Ball Grid Array (9AW) - 45x45x4.28 mm Body [HB2BGA] With Heat Spreader

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	1.00 BSC		
Contact Pad Spacing	C1		43.00	
Contact Pad Spacing	C2		43.00	
Contact Pad Diameter (X1932)	X			0.55
Contact Pad to Contact Pad	G	0.30		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. The theoretically exact value is shown without tolerances.
2. For best soldering results, please refer to the current industry standard IPC-7093.

Microchip Technology Drawing C04-02636 (9AW) Rev B